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ELE 734 - Lab #1 Pre-Lab

(2) Drain Current Equations

NMOS Transistor:

- Cutoff region: $V_{GS} < V_T$

$$I_D = 0$$

- Triode (Linear) Region: $V_{GS} > V_T$ and $V_{DS} < V_{GS} - V_T$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

- Saturation Region: $V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$

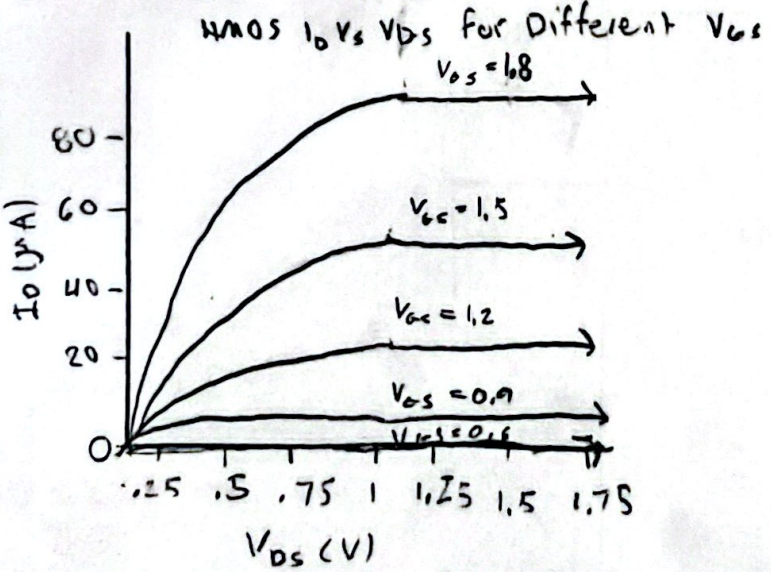
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

PMOS Transistor:

$$I_D = -\mu_p C_{ox} \frac{W}{L} \left[(V_{SG} - |V_{TP}|) V_{SD} - \frac{V_{SD}^2}{2} \right] \text{ (Triode)}$$

$$I_D = -\frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - |V_{TP}|)^2 \text{ (Sat)}$$

IDS - VDS Curve:



(3) Threshold Voltage Equation:

$$V_T = V_{T0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right)$$

* if V_{SB} increases, V_T increases (harder to turn on)
* if V_{SB} decreases, V_T decreases (easier to turn on)

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(4)

CLM is the variation of effective channel length due to increase in V_{DS} beyond saturation.

→ This causes I to increase slightly in Sat region

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T)^2(1+\lambda V_{DS})$$

λ → channel length modulation parameter

- CLM effect is <u>more</u> pronounced in short-channel devices
- long channel devices → ($\lambda \to 0$)

(5)

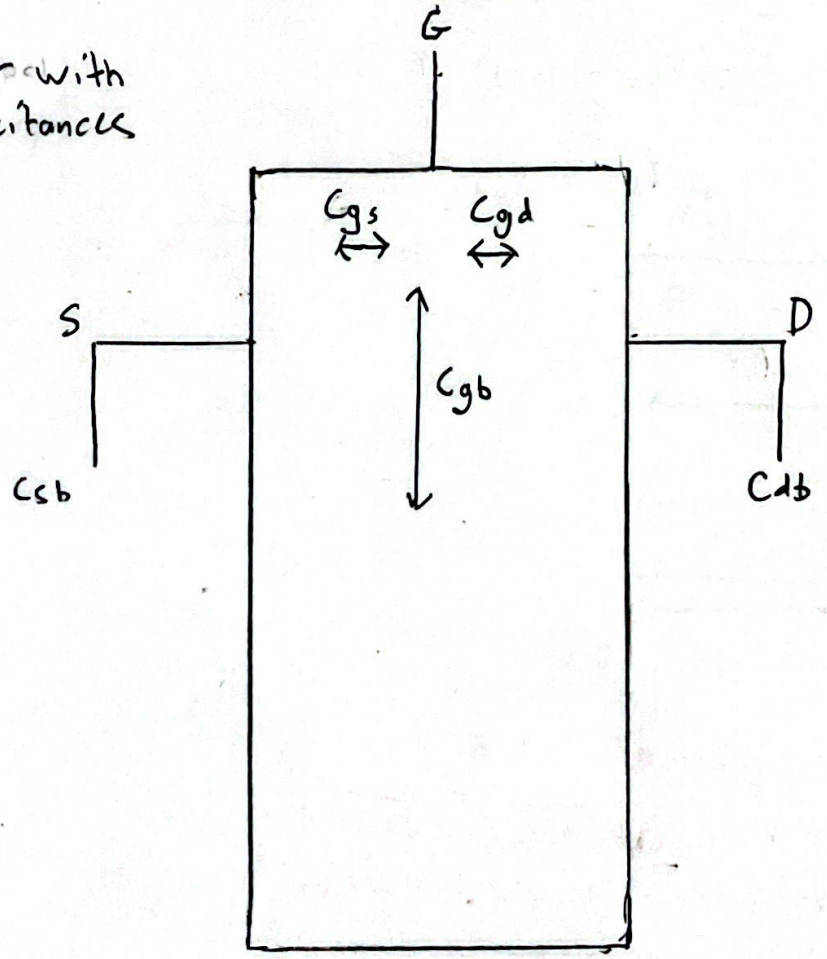
when $V_{GS} < V_T$, MOSFET does not turn off abruptly. Current decreases exponentially:

$$I_D = I_0\, e^{\left(\frac{nV_T}{V_{GS}-V_T}\right)}$$

Sub threshold slope (S):

$$S = n \cdot V_T \cdot \ln(10) \left[\frac{mV}{decade}\right]$$

(6) MOS Transistor with intrinsic capacitances



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